COMP826 Mobile Systems Development

Milestone 2: Prototype, Evaluation and Recommendations

FoodSaver: an AI Food Waste Reduction Assistant

UI/UX prototype, evaluation and design reflection

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1. SYSTEM OVERVIEW

1.1 Project Introduction

Food waste remains a significant environmental and economic problem, with households responsible for around 60% of global food waste (UNEP, 2024). Building on the scoping study submitted for Milestone 1, this project continues the development of FoodSaver, a mobile application that helps users track their food inventory, receive expiry reminders, and get recipe suggestions built around ingredients that are close to their use-by date.

In Milestone 1 the intention was to take on the UI/UX Designer role for this project, and that decision has carried through unchanged into Milestone 2. Rather than building the functional backend and mobile codebase described in the technology stack, this milestone focuses on producing a complete, navigable interface prototype covering the main functional areas identified in Milestone 1, and on evaluating that interface through a timed task baseline and a heuristic evaluation.

The Milestone 1 report received feedback that the proposed navigation diagram only showed one-directional flow, with no clear way for a user to return to the Home screen from deeper parts of the app. That gap has been treated as a priority for this milestone: every screen other than Login and Home includes a route back to the previous screen or to Home, and this is discussed further in Section 3.1.

1.2 Project Deliverables

The interactive prototype is hosted on Figma and can be opened directly, without a Figma account, using the link below:

https://www.figma.com/proto/CzfvzzRniiLk5hpPLPFQn8/AI-Food-Waste-Reduction-Assistant-%E2%80%94-UI-UX

A recorded walkthrough of the prototype, the exported screens, the evaluation results, the project README and this report are available in the GitHub repository below. The walkthrough video is in the evaluation folder.

https://github.com/pheaktraly-netizen/ai-food-waste-assistant

Instructions for opening and navigating the prototype are provided in the repository README.md file.

2. FEATURES EVIDENCE AND DEMO

2.1 Application Flow

The prototype extends the application flow proposed in Milestone 1 with two additional screens: a Forgot Password screen, allowing a user to recover their account without creating a new one, and a dedicated Item Detail screen, which the Milestone 1 use case "View Expiry Information" needed but the Milestone 1 navigation diagram did not include. The updated flow is shown in Figure 1. The main screens follow a hub-and-spoke structure with Home at the centre: Inventory, Recipes, Shopping List, Waste Statistics and Profile each return to Home, while Item Detail and Add Food return to Inventory and the account screens return to Login. This directly addresses the reviewer feedback on Milestone 1, which noted that the original navigation diagram was unidirectional.

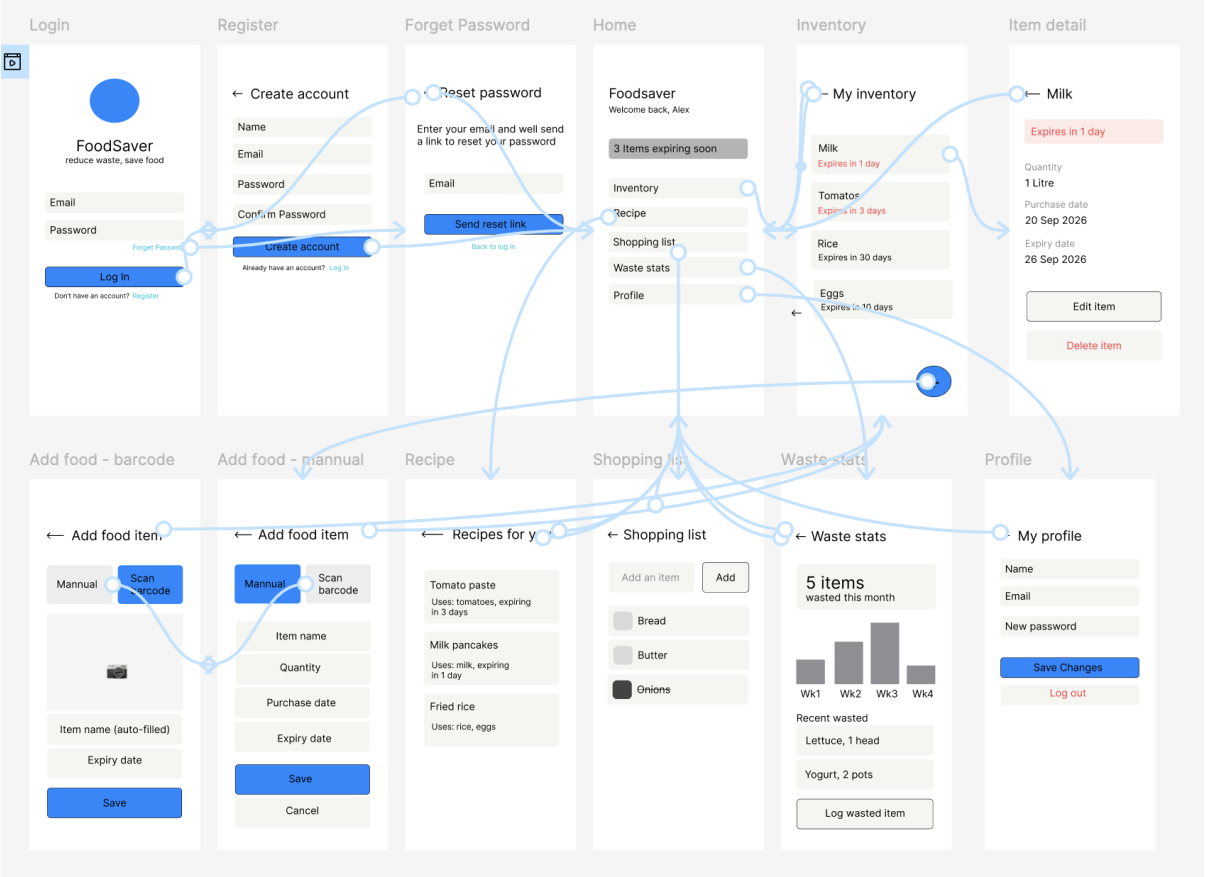


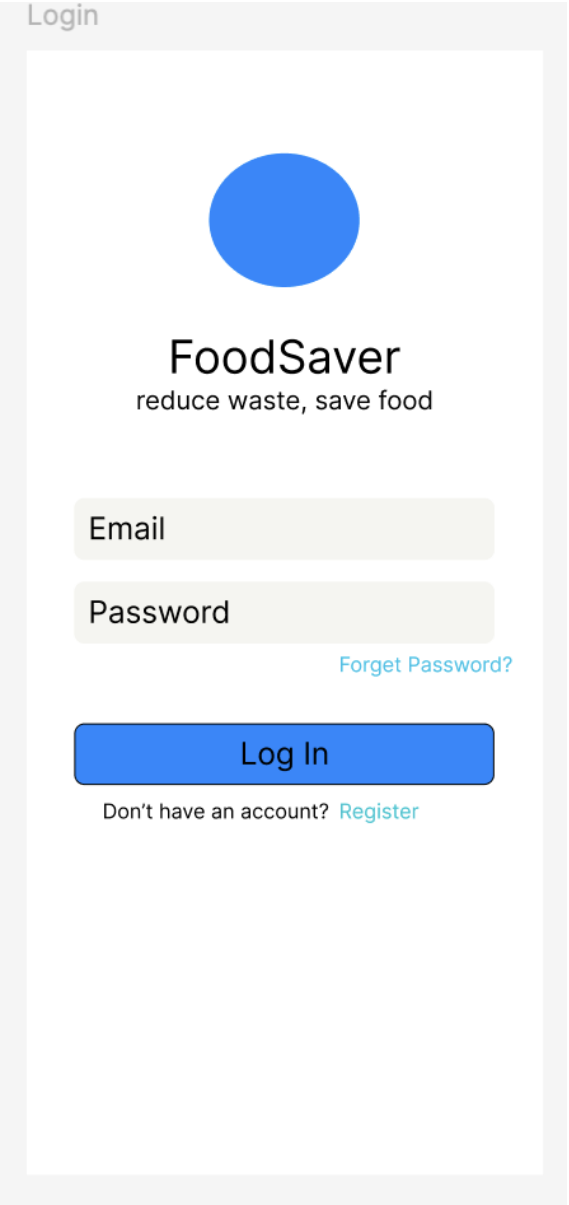
Figure 1. Updated navigation flow for FoodSaver

2.2 Screens and Design Rationale

2.2.1 Login

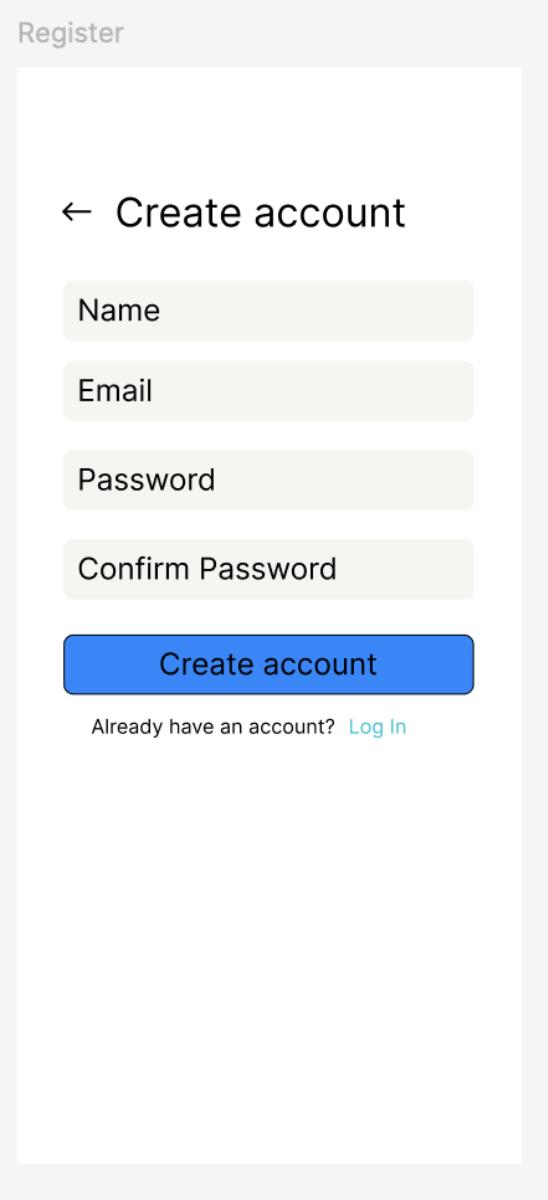
The login screen presents email and password fields, a primary action button, and secondary links to Register and Forgot Password. The two secondary actions are styled

in the accent colour so that they follow the familiar convention that tappable text looks different from static text (Nielsen, 1994). In the coded build the password field would mask typed characters; the prototype shows placeholder text only.



2.2.2 Register

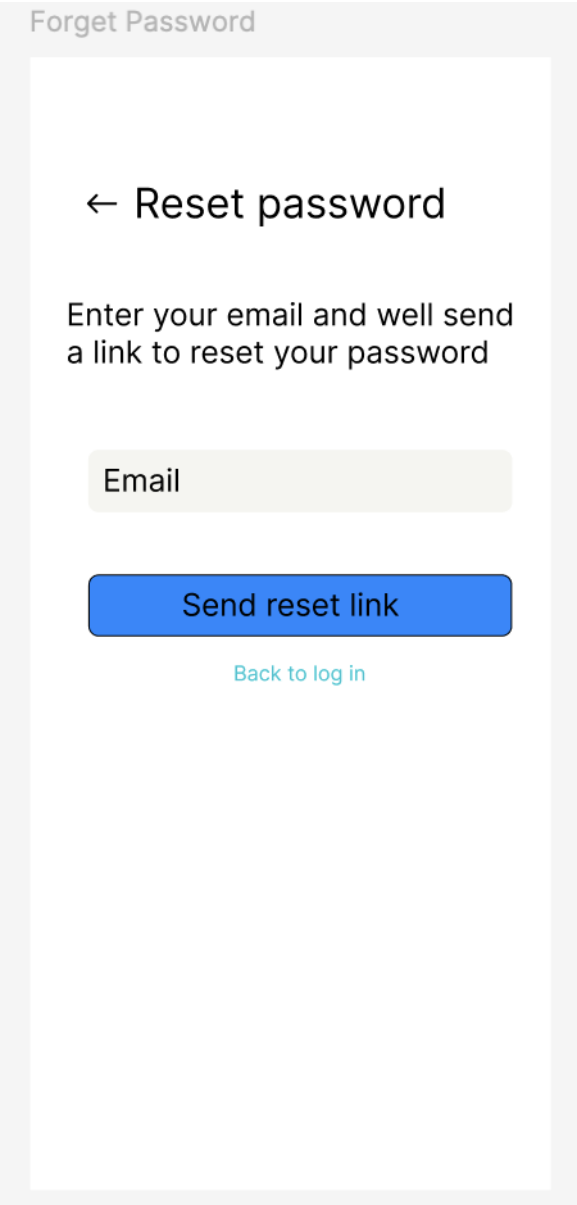
The registration screen collects name, email, password and a password confirmation, reducing the likelihood of a mistyped password going unnoticed. A back arrow and a "Log In" link both return the user to the Login screen, so no screen in the account-creation flow is a dead end.



2.2.3 Forgot Password

This screen was added in Milestone 2 to give users a way to recover access to their account without creating a duplicate one. It follows the same layout pattern as Login and Register so that the recovery flow feels like a natural extension of the existing

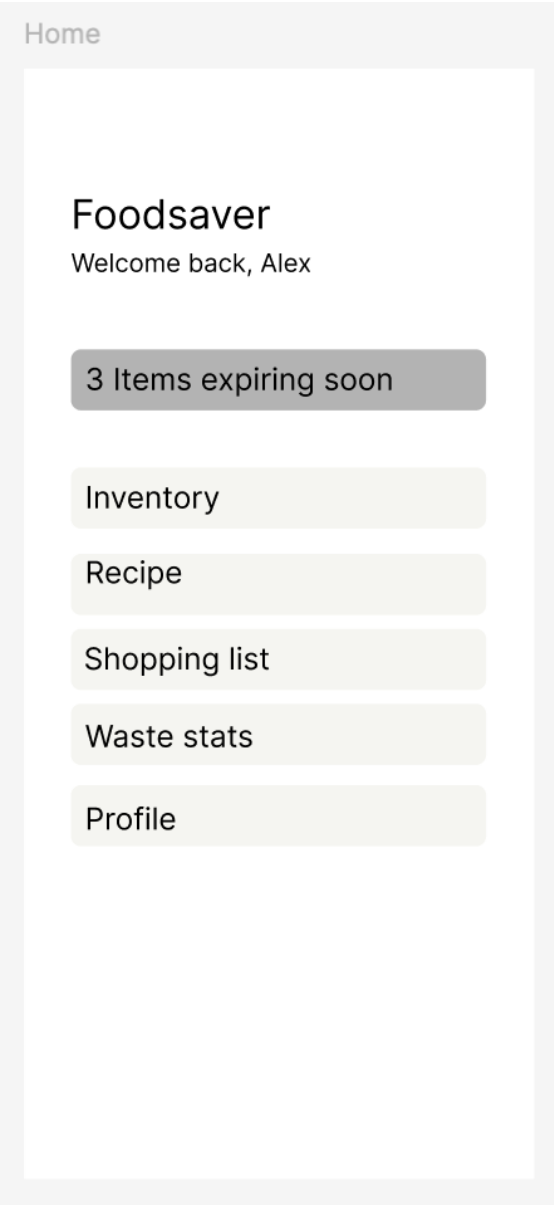
screens rather than a separate system.



2.2.4 Home

Home is the central hub of the application. It greets the user by name, surfaces a summary of items nearing expiry, and provides direct access to the five main functional areas: Inventory, Recipes, Shopping List, Waste Statistics and Profile. Surfacing the expiry summary on the first screen a user sees was a deliberate decision, since prompting users toward food that is about to expire is the core value proposition

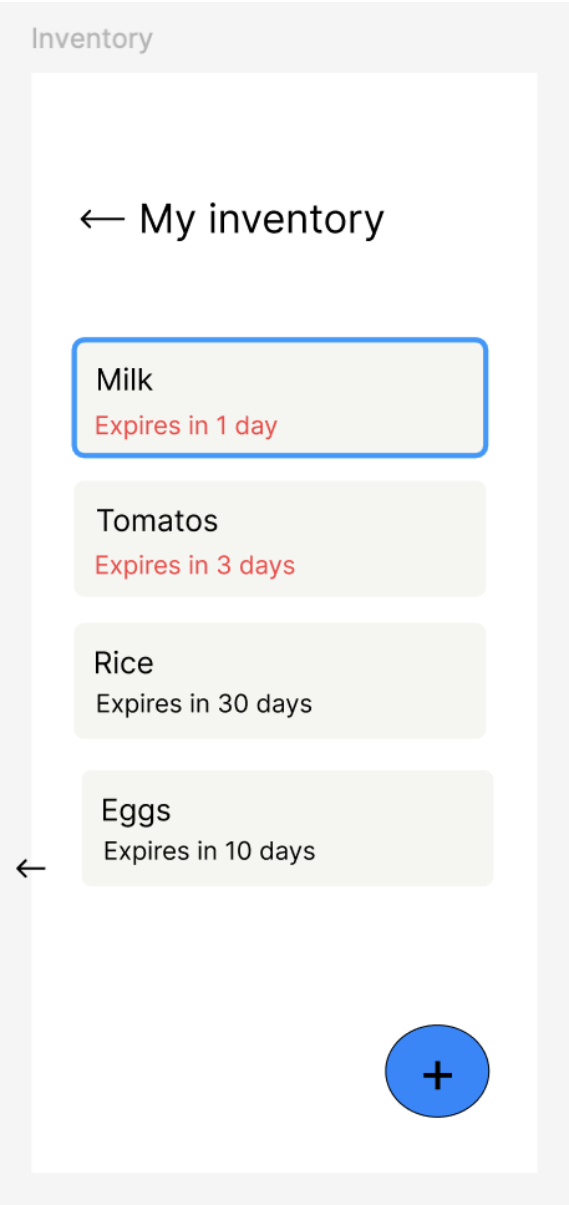
identified in Milestone 1.



2.2.5 Inventory

The inventory screen lists the food items the user has logged, each showing its name and a colour-coded expiry status: red for items that expire within the next few days and neutral grey for items with longer to go. This colour coding lets a user judge urgency without reading every line of text, consistent with the visibility of system status heuristic (Nielsen, 1994). A floating action button in the bottom corner opens the Add Food

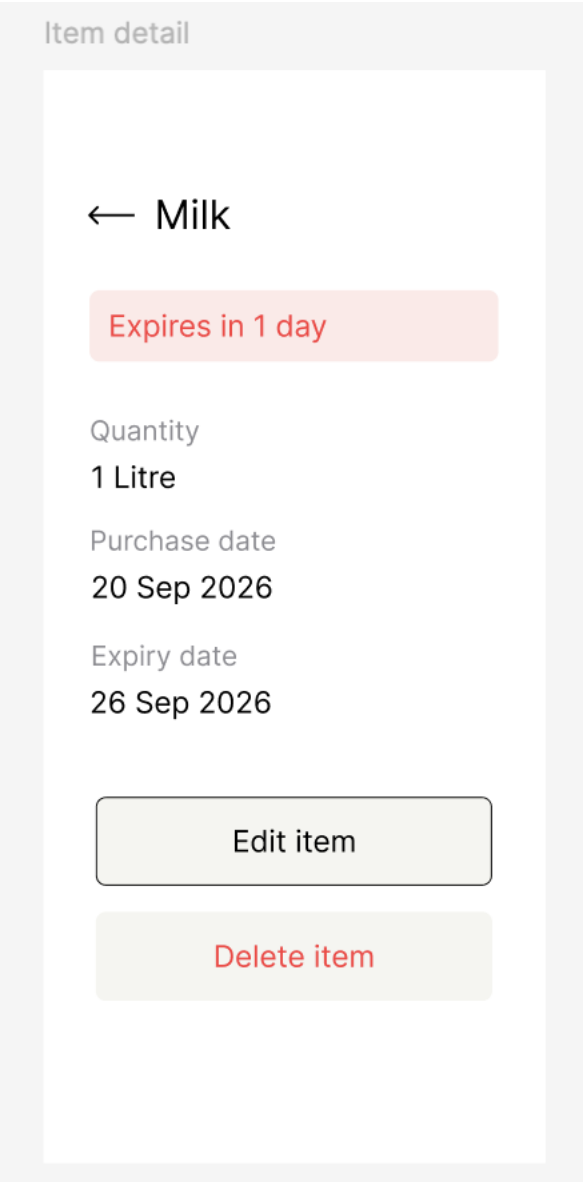
screen.



2.2.6 Item Detail

This screen was introduced to close a gap between the Milestone 1 use case "View Expiry Information" and the proposed screens, where no screen let a user view full expiry information for a single item. It shows quantity, purchase date and expiry date in

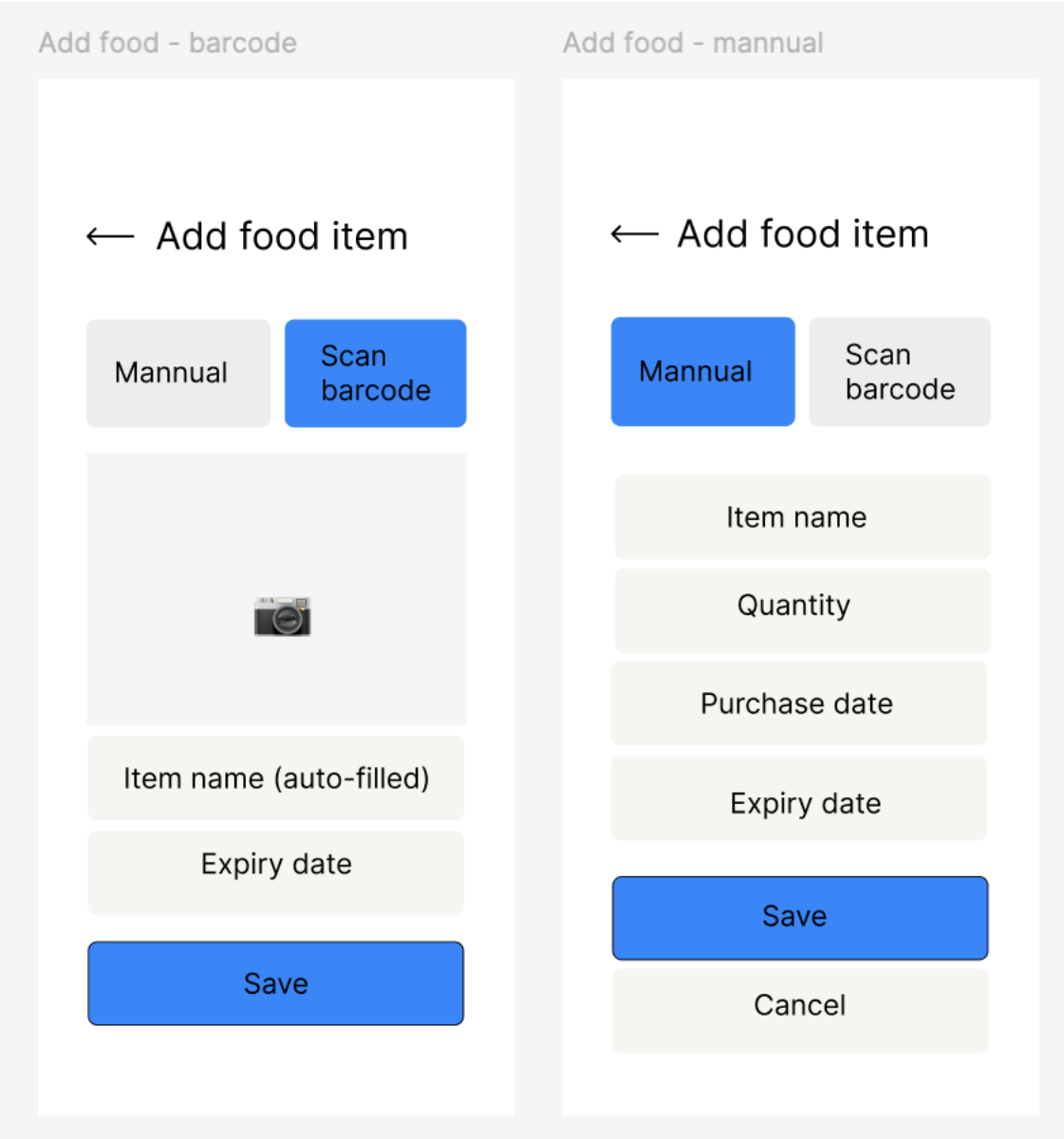
full, alongside Edit and Delete actions.



2.2.7 Add Food (Manual and Barcode)

Adding a food item is supported through two entry modes, switched with a toggle at the top of the screen. The manual mode presents four fields for name, quantity, purchase date and expiry date. The barcode mode represents a camera viewfinder and pre-fills the item name once a barcode is detected, leaving only expiry date for the user to confirm. Offering both paths reflects the Milestone 1 functional requirement that users

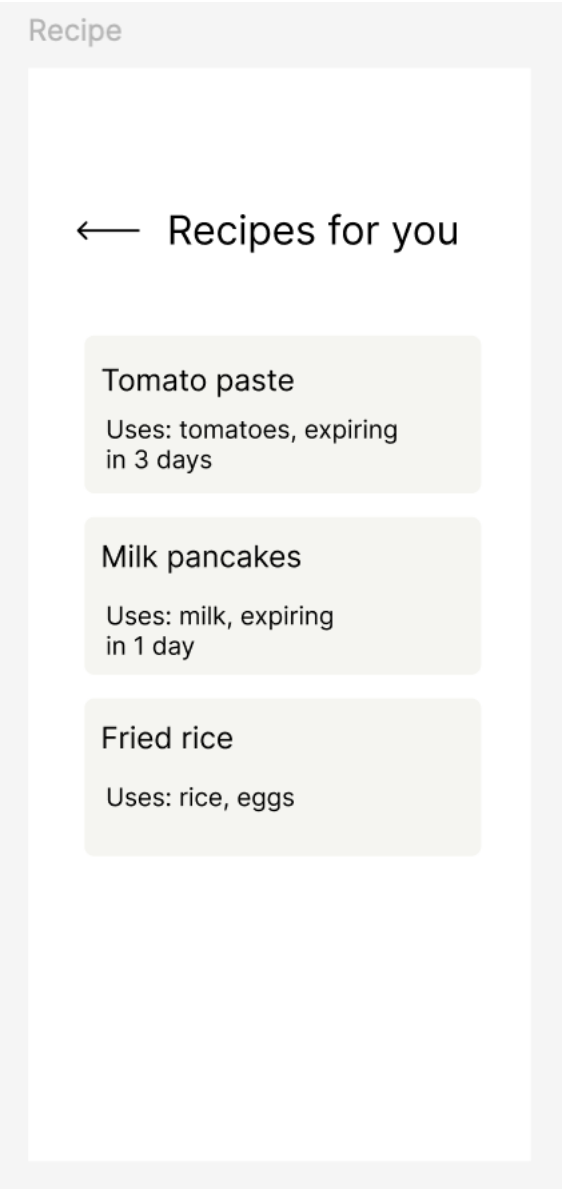
be able to add food manually or via barcode scan.



2.2.8 Recipes

The recipe screen lists recommended recipes as cards, each naming the expiring ingredient the recipe makes use of. This was designed to make the connection between the user's inventory and the recommendation visible at a glance, supporting the Milestone 1 requirement that recipes be recommended based on ingredients the user

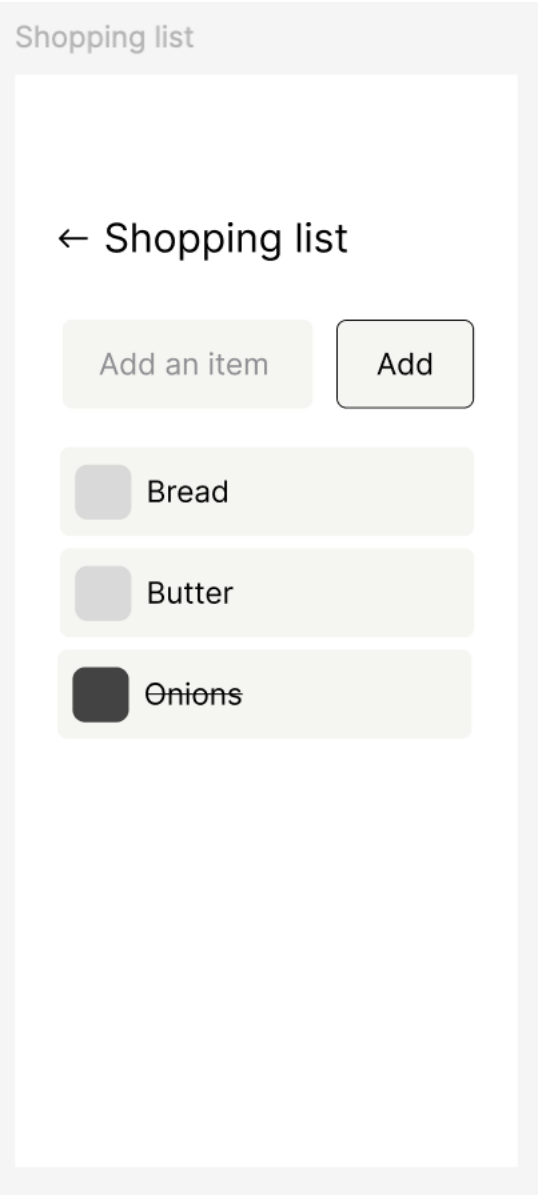
already holds.



2.2.9 Shopping List

Users can add new items to a running shopping list and mark items as purchased. A purchased item is shown with a struck-through label and a filled checkbox rather than being removed outright, so the user retains a short record of what has recently been

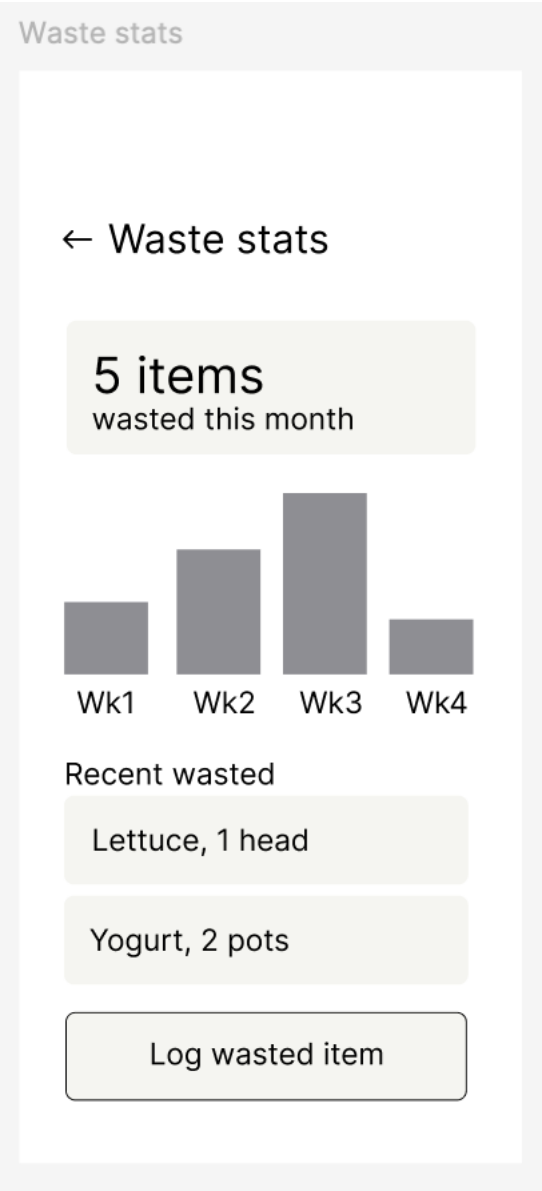
bought.



2.2.10 Waste Statistics

This screen summarises the number of items wasted in the current month, a simple bar chart of waste by week, and a short list of recently wasted items. A "Log wasted item"

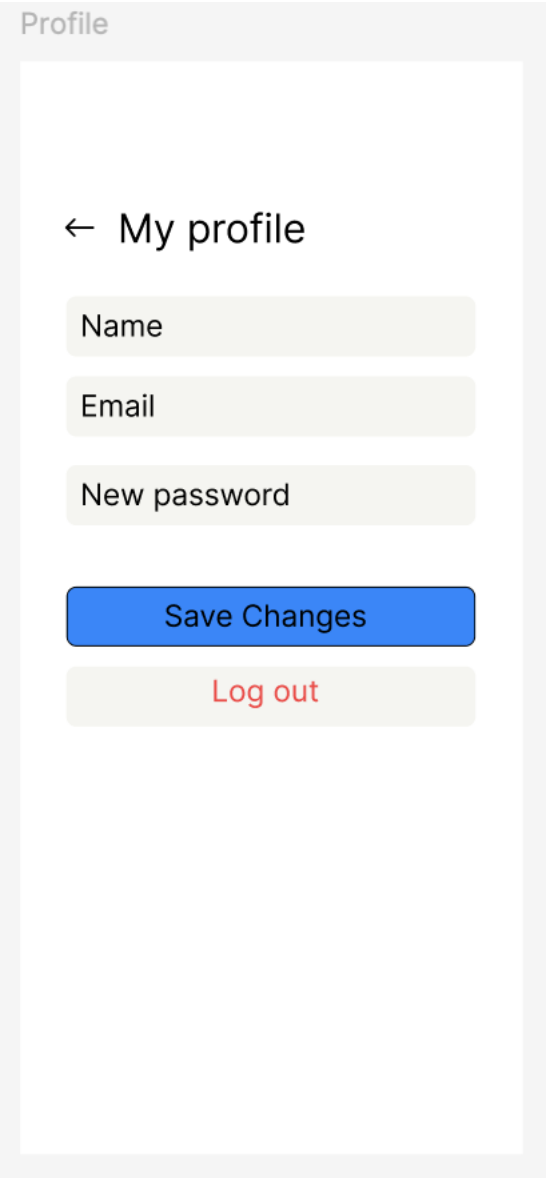
action allows the user to record food that has gone to waste.



2.2.11 Profile

The profile screen allows the user to update their name, email and password, and provides a Log Out action. It follows the same field and button styling used throughout

the rest of the application.



2.3 Known Limitations in the Current Prototype

One interaction remains incomplete in the current build: the "Log wasted item" action on the Waste Statistics screen does not yet lead to a destination screen. Given the scope of the milestone and the time available, priority was given to completing and testing the core account, inventory, recipe and navigation flows, including the add-food path used for the task baseline in Section 4.2, rather than this secondary action. This is discussed further as a limitation in Section 5.2.

3. REFLECTION ON DESIGN AND IMPLEMENTATION

3.1 Design and Implementation

The prototype was built entirely in Figma, using its native prototyping mode to link screens together into a clickable flow rather than writing application code. This decision follows directly from the UI/UX Designer role selected in Milestone 1: the goal of this milestone was to produce and evaluate a complete interface, not to implement the Flutter and Firebase technology stack proposed for the functional build (Figma, n.d.).

A small, consistent design system was used across all eleven screens rather than styling each screen independently. Primary actions, such as Log In, Save and Create Account, use a solid blue fill, while secondary actions use an outlined style. Input fields share a single rounded, light-grey style throughout the application. Reusing the same components across screens, rather than designing each one from scratch, follows the consistency and standards heuristic, which holds that users should not have to wonder whether different words, situations, or actions mean the same thing (Nielsen, 1994).

Colour was also used to carry meaning rather than purely for decoration. Expiry status on the Inventory and Item Detail screens is shown in red for urgent items and grey for the rest, and destructive actions such as Delete and Log Out are shown in red text. This applies the visibility of system status heuristic, which holds that users should be kept informed about what is happening (Nielsen, 1994). The application was named FoodSaver, a straightforward, descriptive name chosen so that a new user can infer the app's purpose before opening it.

A specific implementation issue arose around the iOS status bar and notch area, which initially overlapped the header text on several screens. All header content was moved down by a consistent margin to sit clear of this region.

The most significant technical difficulty encountered was making the back-navigation links reliable inside Figma's prototyping tool. The back arrow icon used on each screen is drawn as a thin vector line with effectively no clickable area, which meant that dragging a connection directly from the arrow produced a link that could not reliably be tapped in preview mode. This was resolved by placing an invisible rectangle over each header, sized generously around the arrow and title text, and attaching the navigation link to that rectangle instead. This is a common workaround in Figma prototyping and reflects a broader lesson about interactive prototyping tools: a link that is technically present in the design file is not the same as a link that is usable in practice, and every interaction has to be tested, not just built.

3.2 Development Process

Milestone 1 proposed a hybrid Scrumban approach for this project, combining the short, structured cycles of Scrum with the continuous, visual workflow of Kanban (Ozkan et al., 2022). This mix was chosen because it gives structure for planning while keeping progress easy to monitor, and a board is a widely used way to make work visible (Ahmad et al., 2013). For Milestone 2, it was carried through using a Trello board with To Do, In Progress, Testing and Done columns, mirroring the structure set out in the Milestone 1 process diagram. The board is shown in Figure 2.

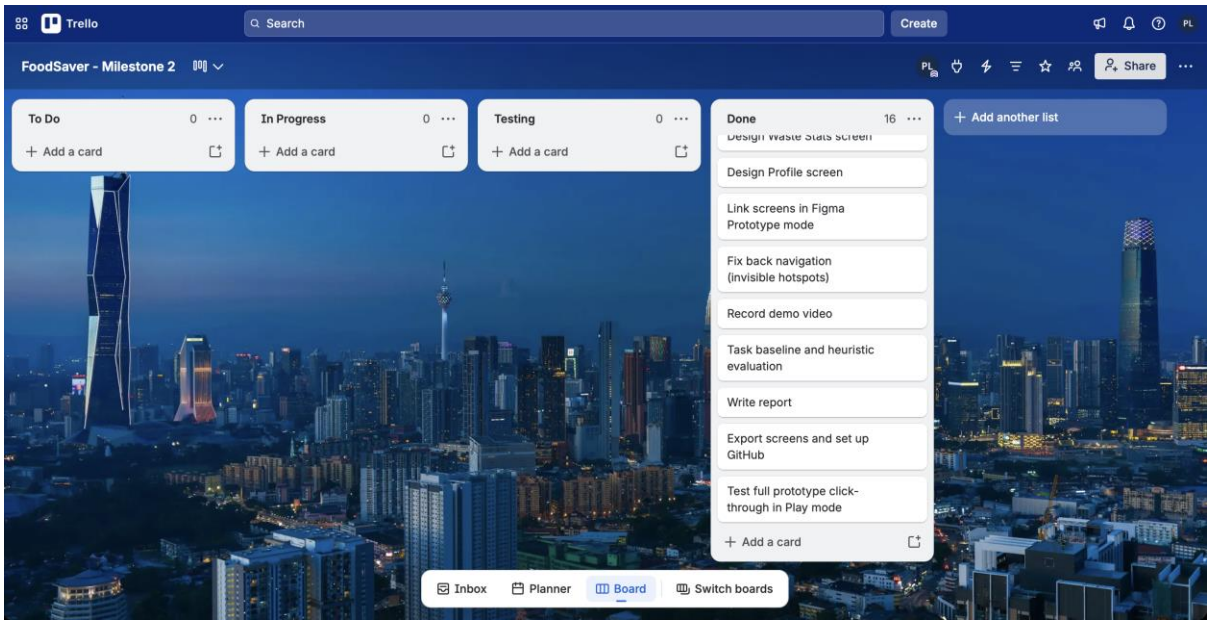


Figure 2. Trello board used to track screen production

The board was set up after the main screens had been designed, so the initial design work was recorded retrospectively, while the remaining tasks (testing, demo recording, the evaluation and report writing) were tracked live as they moved across the columns. Screens were grouped into cards by function, for example the two Add Food screens on one card, rather than one card per frame, which kept the board readable for a solo project. In practice, the Kanban half of the Scrumban approach proved more useful than the Scrum half for a solo UI/UX task of this kind. Because each screen could be designed, reviewed and tested largely independently of the others, a steady flow of cards to Done suited the work better than fixed-length sprints with a formal planning and review cycle at each end, which is consistent with the flexibility associated with Kanban's continuous flow (Ahmad et al., 2013).

The clearest process lesson was that testing needed to be its own stage rather than assumed once a screen looked finished. Several screens that appeared complete in the Design tab had missing or broken links when tested in Figma's Prototype preview, which is the kind of problem a dedicated Testing column is meant to catch before a card is

moved to Done. Setting up the board at the start, rather than after the main screens, would also have given a real record of how long each screen took.

4. SYSTEM EVALUATION

Evaluation followed the approach set out in Milestone 1, which used the PACMAD usability model. PACMAD extends the three ISO 9241-11 attributes of effectiveness, efficiency and satisfaction with learnability, memorability, errors and cognitive load (Harrison et al., 2013). The evaluation was carried out by the designer using two methods: a timed run of the add-food task, which gives a best-case baseline for effectiveness and efficiency, and a heuristic evaluation against Nielsen's ten heuristics (Nielsen, 1994). Satisfaction, learnability, memorability, errors and cognitive load depend on first-time users, so they were not measured and are left for future testing.

Milestone 1 set six evaluation criteria: usability, performance, reliability, security, portability and functional correctness. Usability can be evaluated directly on a prototype, and basic navigation correctness was checked through the click-through testing described in Section 3.2. Performance, reliability, security and portability depend on a working app with a database, authentication and real devices, so they are deferred to the coded build.

4.1 How Mobile Systems Are Evaluated

Mobile usability is evaluated through a mix of experimental, survey and inspection methods. In the 163 studies Harrison et al. (2013) reviewed from 2008 to 2010, controlled experiments were the most common method (about 60%), followed by field studies (about 27%) and surveys (about 4%), while expert evaluation accounted for only around 1%. Weichbroth's (2020) later review found efficiency (70%), satisfaction (66%) and effectiveness (58%) were the most frequently measured attributes, with controlled observation and surveys the two main methods. Practitioner guidance from the Nielsen Norman Group describes heuristic evaluation as a method for identifying design problems in a user interface (Moran & Gordon, 2023). Two methods were therefore applied: a task test and a heuristic evaluation, covering behaviour and expert inspection.

4.2 Task Test: Designer Baseline

The Milestone 1 target, that 80% of users can add a food item within one minute, cannot be tested properly without users who have not seen the design. As a best-case baseline, the author completed the add-food task three times, starting from Home and timing from the first tap on Inventory to the tap on Save. All three runs were far inside the one-minute limit, and the first was the slowest, which suggests that even the

designer needed a moment to settle into the path. Because the author designed the screens, this shows that the path is short and works, but not whether a new user would find it, so the Milestone 1 target remains untested.

Run	Time (seconds)	Taps	Completed?
1	15	3	Yes
2	6	3	Yes
3	9	3	Yes
Mean	10	3	3 of 3

Table 1. Designer baseline for the add-food task (shortest path: Inventory, the add button, Save)

4.3 Heuristic Evaluation

Heuristic evaluation is a method in which an evaluator checks an interface against recognised usability principles rather than observing users (Moran & Gordon, 2023). Nielsen's ten usability heuristics were used as the checklist (Nielsen, 1994). It cannot replace testing with real users, so it was used here alongside the task baseline. Severity was rated low (cosmetic), medium (causes hesitation or a dead end) or high (blocks a task). The problems found are recorded in Table 2.

Screen	Heuristic	Problem found	Severity	Action
Login, Reset password	Match between system and real world	The link on Login reads "Forget Password?", and the Reset password text reads "well send" instead of "we'll send".	Low	Not corrected.
Home	Consistency and standards	The app name appears as "Foodsaver" where Login shows "FoodSaver", and the navigation button reads "Recipe" while its screen is titled "Recipes for you".	Low	Not corrected.
Add Food	Match between system and real world	The entry-mode toggle reads "Mannual" instead of "Manual".	Low	Not corrected.
Waste Statistics	Visibility of system status	Tapping "Log wasted item" gives no response because no destination screen exists.	Medium	Not corrected; recorded as a known limitation (Section 2.3).
Item Detail	Error prevention	Delete item acts immediately with no confirmation step, so an	Medium	Not corrected in the prototype;

		accidental tap could remove an item.		recommended for the coded build.
Add Food (barcode)	Visibility of system status	The barcode mode shows only a camera icon, with no indication of scanning progress or of what happens if a barcode is not recognised.	Low	Not corrected; recommended for the coded build.

Table 2. Heuristic evaluation findings

4.4 Discussion and Reflection

The baseline runs completed in 6 to 15 seconds, always with the minimum three taps, so the add-food path is short. Because the designer ran them, this shows the path works, not that a new user would find it, so the Milestone 1 target of 80% of users within one minute remains untested. The heuristic evaluation recorded six findings, none of which were corrected in the prototype. Three are cosmetic wording and consistency problems; the others are the unfinished "Log wasted item" action and design gaps (no delete confirmation, no scanning feedback) that a coded build would need to address. The first changes would be a confirmation step for Delete item, a visible scanning state on the barcode screen, and a destination for "Log wasted item". The main limitation is that one evaluator, who is also the designer, carried out both methods, so problems the designer cannot see are likely to have been missed. Testing with first-time users, and a second evaluator running the heuristic checklist independently, would give a more dependable picture.

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

This milestone delivered a complete, navigable prototype of the FoodSaver application covering the main functional areas identified in Milestone 1, and addressed the navigation weakness raised in the Milestone 1 feedback by ensuring every screen after login provides a route back. Building the prototype in Figma, rather than as working code, allowed a consistent design system, colour-coded urgency states, and a full click-through flow to be produced and tested within the time available. The timed baseline showed that adding a food item takes three taps and 6 to 15 seconds, and the heuristic evaluation recorded six findings, mostly cosmetic. The main work left is testing with first-time users and building the functional app behind the interface.

5.2 Limitations

- No functional backend, database or barcode scanning logic exists behind the prototype; all data shown is static example content.
- The "Log wasted item" action on the Waste Statistics screen is not linked to a completed destination screen.
- The prototype fields are static and accept no typed input, so the task baseline timed navigation rather than data entry.
- No external participants took part, so efficiency, effectiveness, satisfaction and learnability for new users remain untested, and the heuristic evaluation was carried out by one evaluator who is also the designer, so problems the designer cannot see are likely to have been missed.
- The Milestone 1 criteria for performance, reliability, security and portability could not be tested on a click-through prototype, and real-world constraints such as network latency, camera permissions and notification delivery were also untested.

5.3 Future Work

Next step. Based on this experience, the most direct next step for FoodSaver would be implementing the technology stack proposed in Milestone 1, using the Figma prototype as a visual specification for a Flutter build. Figma's inspect and developer handoff features, which expose spacing, colour and typography values, should reduce the gap between the design produced in this milestone and a working implementation.

Process. This project showed how far a prototyping tool can go before any code exists: a clickable Figma flow was enough to test navigation and the add-food path. Commercial teams could adapt this by prototyping and testing every flow before development starts, and by keeping a shared component library rather than rebuilding styles per screen. A design system would likely have reduced the rework of refining button and field styles across eleven separate screens in this project.

Tooling. Two gaps affected this project. First, the form fields in the prototype could not accept typed input, so the add-food task tested navigation but not data entry; prototyping tools that allow real text entry would make timed tasks more realistic. Second, an automated usability-testing service that records unmoderated click-through sessions and reports where users drop off would have been more useful than timed runs by the designer alone. Such a service would let far more people test a prototype before any code is written, and would capture timing and tap data without the researcher recording it by hand.

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